Web and Mobile Application Development

Khaula Molapo

20240001

Week 7

1.

An Android e-commerce application allows users to buy products using their mobile phones. The app is connected to an online store system where products, prices, and customer information are stored in a database. First, the developer creates the project in Android Studio and designs the user interface. The interface normally includes pages such as login, product list, product details, shopping cart, and payment page.

The application communicates with a backend server using APIs. These APIs allow the app to request data such as available products, prices, and customer orders. When a user logs in, the system verifies their details from the server database. Users can then browse products, add items to the cart, and proceed to checkout.

Security is also important. The app should use authentication and encrypted connections to protect user information. Notifications can also be added to inform users about discounts, order status, or new products. This setup helps businesses sell products easily while giving customers a simple and convenient shopping experience.

2.

Android activities are one of the most important components in Android application development. An activity represents a single screen in an Android application where the user can interact with the system. For example, a login screen, product page, or settings page are all examples of activities.

Each activity controls the user interface and handles user actions such as button clicks or text input. Activities follow a lifecycle which includes stages like creation, starting, pausing, and stopping. These stages help the Android system manage memory and performance efficiently.

Activities also allow developers to move between screens using something called Intents. For example, when a user presses a button to view product details, the app moves from the product list activity to the product details activity. This makes applications organized and easy to navigate. Activities therefore help structure the app and improve user experience.

3.

Android Studio is the main tool used to develop Android applications. One important feature of Android Studio is the UI Layout Editor, which helps developers design the user

interface of an application. In this tool, developers can drag and drop components such as buttons, text boxes, images, and menus onto the screen.

The layout editor usually uses XML files to define how the interface looks. Developers can choose layouts such as LinearLayout, ConstraintLayout, or RelativeLayout to arrange components properly. This makes the design flexible and easy to manage.

Creating a UI layout in Android Studio helps developers see how the application will look on different screen sizes. It also improves the user experience because the interface can be designed to be clear and easy to use. Overall, the layout tool helps developers build clean and professional-looking mobile applications more efficiently.

4.

Android app architecture refers to the structure used to design and organize an Android application. A good architecture helps developers manage code, improve performance, and make the app easier to maintain. One common architecture used in Android development is the Model-View-ViewModel (MVVM) architecture.

In this structure, the Model handles the data and business logic of the application. The View represents the user interface that the user interacts with. The ViewModel acts as a connection between the Model and the View. It manages the data that the UI needs to display.

This separation makes the code cleaner and easier to test. If changes are made in one part of the system, it does not break the entire application. Using proper architecture also helps developers work in teams more effectively and build applications that are stable, scalable, and easier to update in the future.